

Trainings ▾

General Information

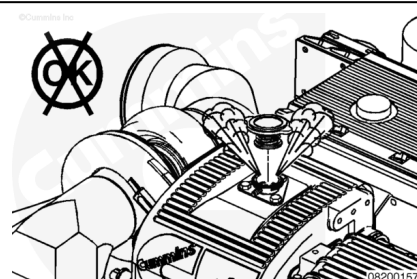
Restore™ is a heavy-duty cooling system cleaner that removes corrosion, silica gel, and other deposits. The performance of Restore™ is dependent on time, temperature, and concentration levels. An extremely scaled or flow-restricted system, for example, can require higher concentrations of cleaners, higher temperatures, or longer cleaning times, or the use of Restore Plus™. Up to twice the recommended concentration levels of Restore™ can be used safely. Restore Plus™ **must** be used **only** at its recommended concentration level. Extremely scaled or fouled systems can require more than one cleaning.



Drain

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

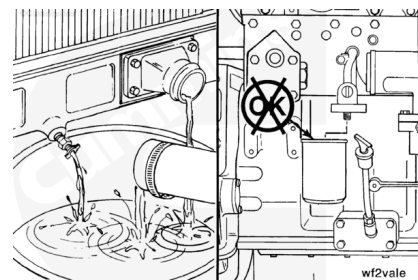
Coolant is toxic. Keep away from children and animals. If not reused, dispose of in accordance with local environmental regulations.

Remove the pressure cap.

Note : Not all engines are assembled with a coolant filter. The use of coolant filter is optional when using coolant meeting Cummins® coolant requirements.

Drain the cooling system. Do **not** allow the cooling system to dry out.

Do **not** remove the coolant filter.

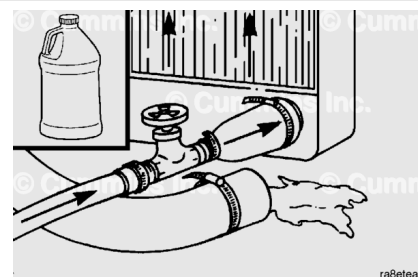


Flush

⚠ CAUTION ⚠

Do not use caustic cleaners in the cooling system. Aluminum components will be damaged.

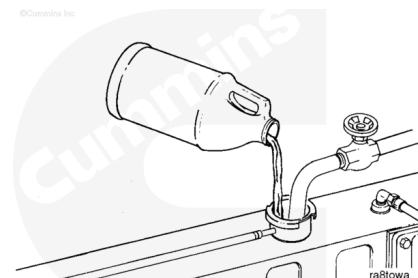
The cooling system **must** be clean to work correctly and to eliminate buildup of harmful chemicals.



⚠ CAUTION ⚠

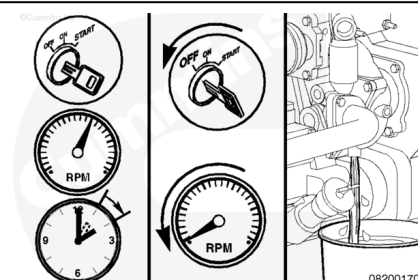
Fleetguard® Restore™ contains no antifreeze. Do not allow the cooling system to freeze during the cleaning operation.

Immediately add 3.8 liters [1 gal] of Fleetguard® Restore™, Restore™ Plus, or the equivalent, for each 38 to 57 liters [10 to 15 gal] of cooling system capacity, and fill the system with plain water.

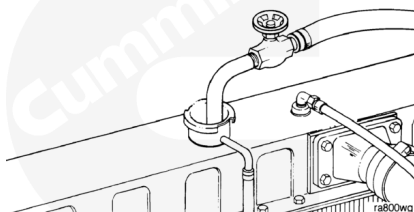


Operate the engine at normal operating temperatures, at least 85°C [185°F], for 1 to 1-1/2 hours.

Shut the engine OFF, allow to cool to 50° C [120°F], and drain the cooling system.

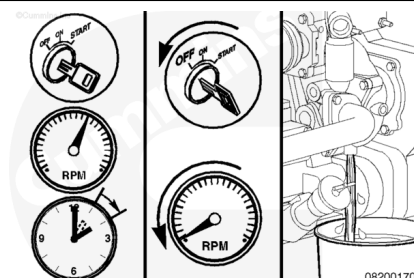


Fill the cooling system with clean water.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Operate the engine at HIGH IDLE for five minutes with the coolant temperature above 85°C [185°F].

Shut the engine OFF, allow to cool to 50° C [120°F] and drain the cooling system.

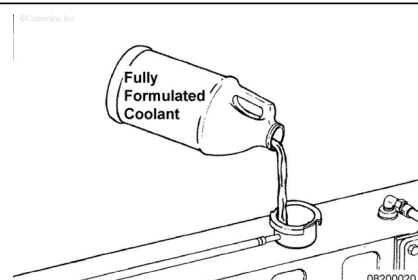
If the water being drained is still dirty, the system **must** be flushed again until the water is clean.

Fill

⚠ CAUTION ⚠

To reduce the possibility of engine damage, do not add coolant while the engine is hot.

Fill the cooling system with fully formulated coolant or a 50/50 mixture of fully formulated antifreeze and good quality water. Use a service filter to bring the coolant to the correct SCA concentration level. Refer to Procedure 018-004 in Section V. (</qs3/pubsys2/xml/en/procedures/101/101-018-004.html>)



Install the pressure cap. Operate the engine until it reaches a temperature of 80°C [180°F], and check for coolant leaks.

The freeze point protection **must** be checked after coolant is added to the cooling system.

The Fleetguard™ refractometer, Part Number C2800, provides a reliable, easy-to-read, and accurate measurement of freeze point protection and antifreeze concentration.

Refer to the OEM service manual instructions for the correct operation instructions for the Fleetguard® refractometer, Part Number C2800.

